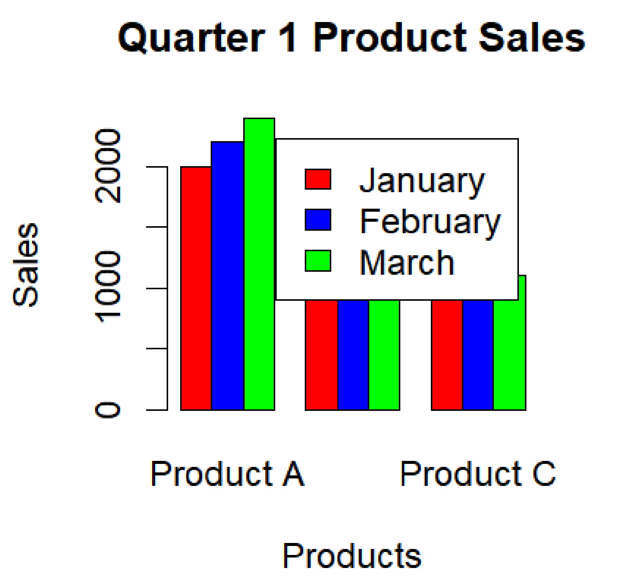
Lab Experiments 6 - 12

SET 6 – Product Sales Analysis   17 – 7 - 26

Q1. Grouped Bar Chart

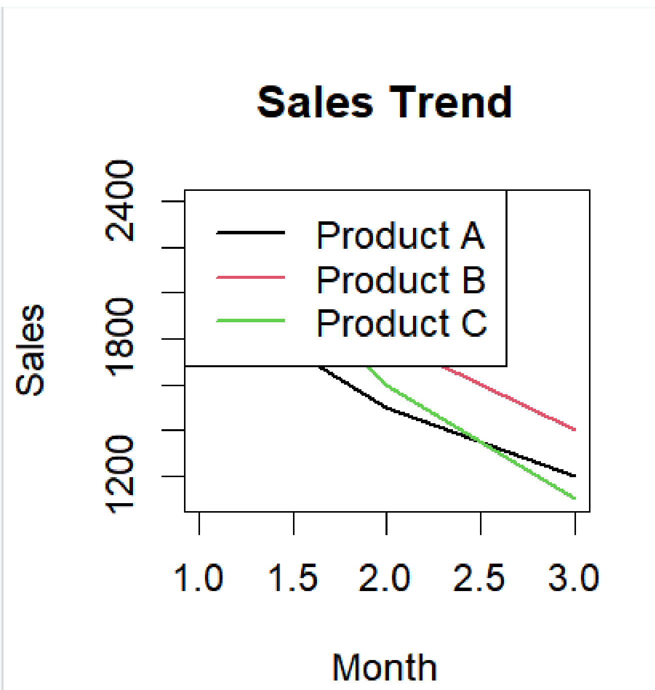
```
# Product Sales Data
Product <- c("Product A","Product B","Product C")

January <- c(2000,1500,1200)
February <- c(2200,1800,1400)
March <- c(2400,1600,1100)

sales <- rbind(January,February,March)

barplot(sales,
    beside=TRUE,
    names.arg=Product,
    col=c("red","blue","green"),
    legend.text=c("January","February","March"),
    xlab="Products",
    ylab="Sales",
    main="Quarter 1 Product Sales")
```



Q2. Stacked Area Chart

```
Product <- c("Product A","Product B","Product C")


January <- c(2000,1500,1200)

February <- c(2200,1800,1400)

March <- c(2400,1600,1100)


Month <- c(1,2,3)


matplot(Month,

      cbind(January,February,March),

      type="l",

      lwd=2,

      lty=1,
```

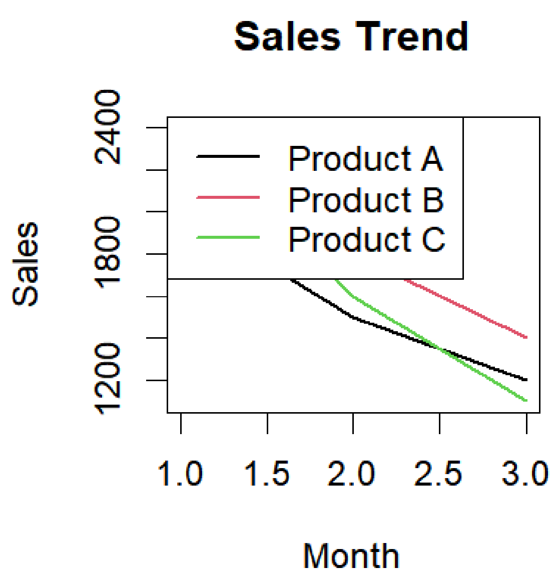
```
       xlab="Month",
       ylab="Sales",
       main="Sales Trend")

legend("topleft",
      legend=Product,
      col=1:3,
      lty=1,
      lwd=2)
```



Q3. Table

```
Product <- c("Product A","Product B","Product C")

January <- c(2000,1500,1200)

February <- c(2200,1800,1400)

March <- c(2400,1600,1100)
```

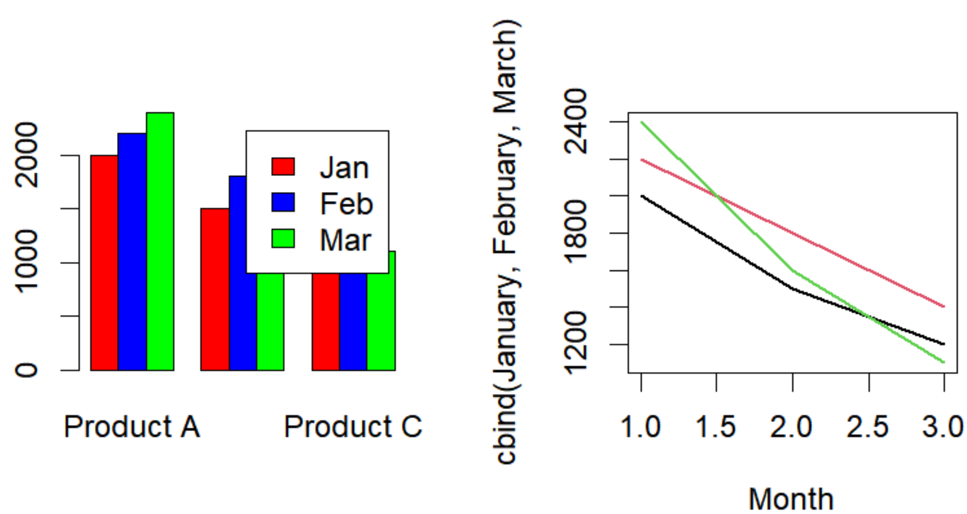
```
sales_table <- data.frame(Product,
                 January,
                 February,
                 March)

print(sales_table)
```



Q4. Dashboard (R Equivalent)

```
par(mfrow=c(1,2))

Product <- c("Product A","Product B","Product C")

January <- c(2000,1500,1200)

February <- c(2200,1800,1400)

March <- c(2400,1600,1100)
```

```
sales <- rbind(January,February,March)

barplot(sales,
        beside=TRUE,
        names.arg=Product,
        col=c("red","blue","green"),
        legend.text=c("Jan","Feb","Mar"))

Month <- c(1,2,3)

matplot(Month,
        cbind(January,February,March),
        type="l",
        lwd=2,
        lty=1)
```



SET 7 – Customer Demographics Analysis

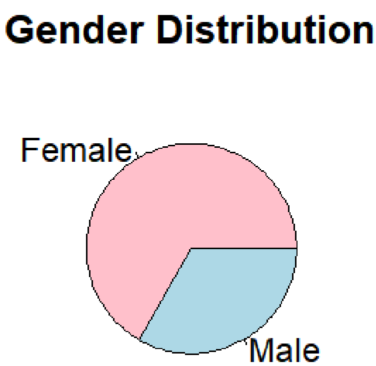
Q1. Bar Chart

```
Customer_ID <- c(1,2,3)

Age <- c(28,35,42)


barplot(Age,
    names.arg=Customer_ID,
    col="skyblue",
    xlab="Customer ID",
    ylab="Age",
    main="Customer Age Distribution")
```

Q2. Pie Chart

```
Gender <- c("Female","Male","Female")
```

```
pie(table(Gender),
   labels=names(table(Gender)),
   col=c("pink","lightblue"),
   main="Gender Distribution")
```



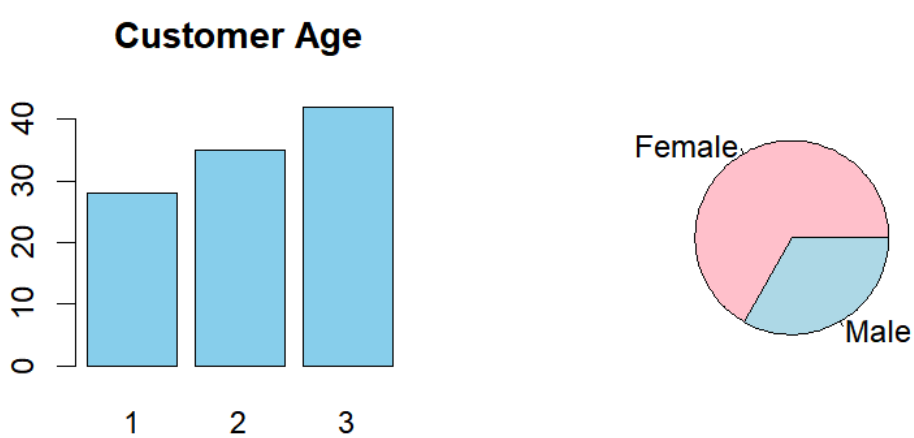
Q3. Table

```
Customer_ID <- c(1,2,3)
Age <- c(28,35,42)
Gender <- c("Female","Male","Female")
Income <- c(50000,60000,75000)
customer_table <- data.frame(Customer_ID,
                     Age,
                     Gender,
                     Income)
print(customer_table)
```

```
Customer_ID Age Gender Income
          1  28 Female  50000
          2  35   Male  60000
          3  42 Female  75000
```

Q4. Dashboard (R Equivalent)

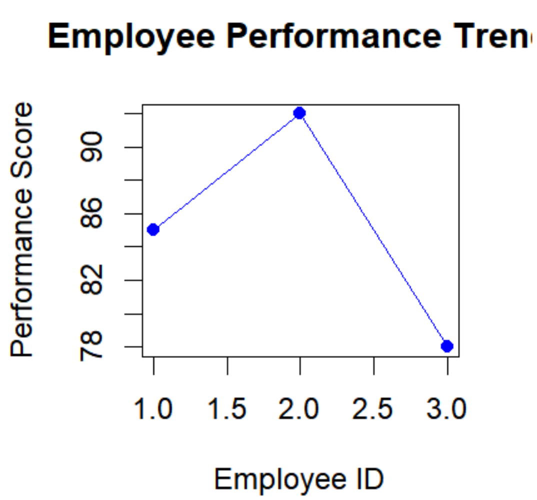
```
par(mfrow=c(1,2))


Customer_ID <- c(1,2,3)

Age <- c(28,35,42)


barplot(Age,

        names.arg=Customer_ID,

        col="skyblue",

        main="Customer Age")


Gender <- c("Female","Male","Female")


pie(table(Gender),

    labels=names(table(Gender)),

    col=c("pink","lightblue"))
```



SET 8 – Employee Performance Analysis

Q1. Line Chart

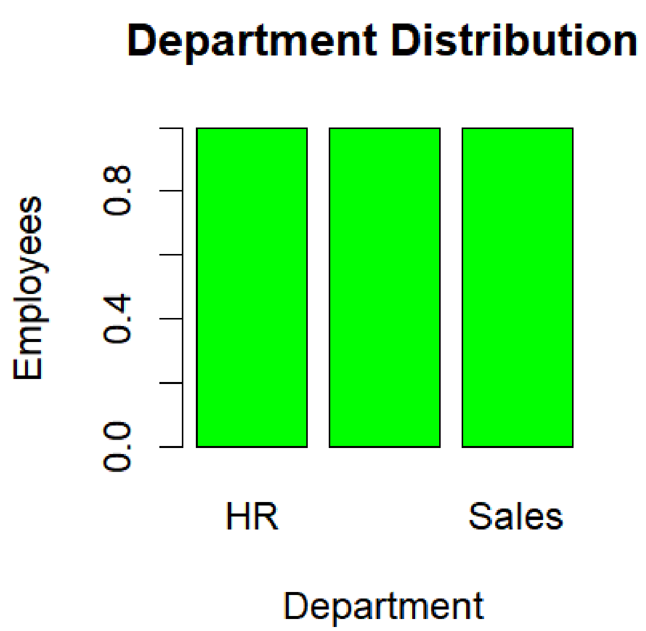
```
Employee_ID <- c(1,2,3)

Performance <- c(85,92,78)


plot(Employee_ID,
    Performance,
    type="o",
    pch=19,
    col="blue",
    xlab="Employee ID",
    ylab="Performance Score",
    main="Employee Performance Trend")
```



Q2. Bar Chart

```
Department <- c("Sales","HR","Marketing")

barplot(table(Department),
      col="green",
```

```
        xlab="Department",
        ylab="Employees",
        main="Department Distribution")
```



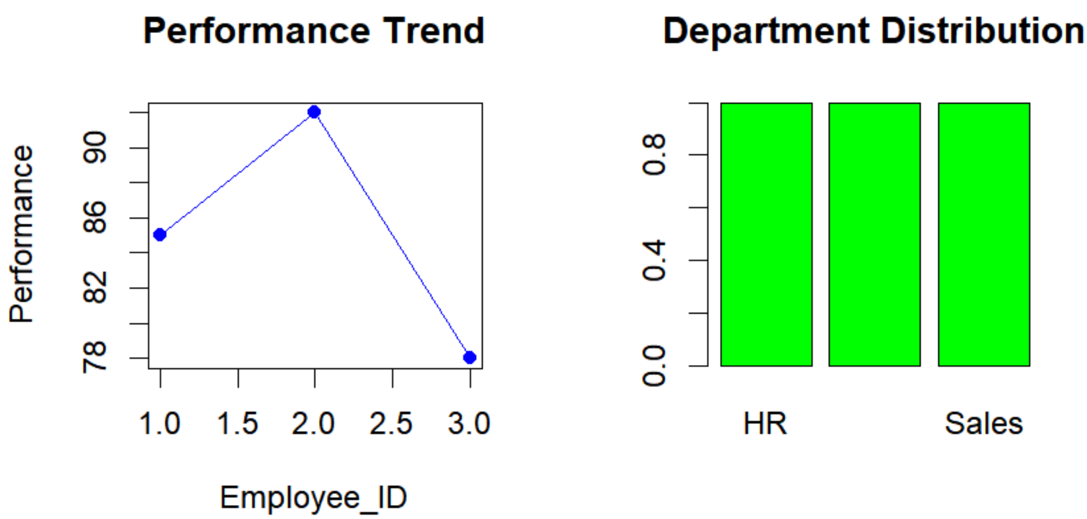
Q3. Table

```
Employee_ID <- c(1,2,3)
Department <- c("Sales","HR","Marketing")
Years_of_Service <- c(5,3,7)
Performance <- c(85,92,78)

employee_table <- data.frame(Employee_ID,
                             Department,
                             Years_of_Service,
                             Performance)

print(employee_table)
```

```
Employee_ID Department Years_of_Service Performance
          1      Sales                5          85
          2         HR                3          92
          3  Marketing                7          78
```

Q4. Dashboard (R Equivalent)

```
par(mfrow=c(1,2))


Employee_ID <- c(1,2,3)

Performance <- c(85,92,78)


plot(Employee_ID,

   Performance,

   type="o",

   pch=19,

   col="blue",

   main="Performance Trend")


Department <- c("Sales","HR","Marketing")


barplot(table(Department),

     col="green",

     main="Department Distribution")
```

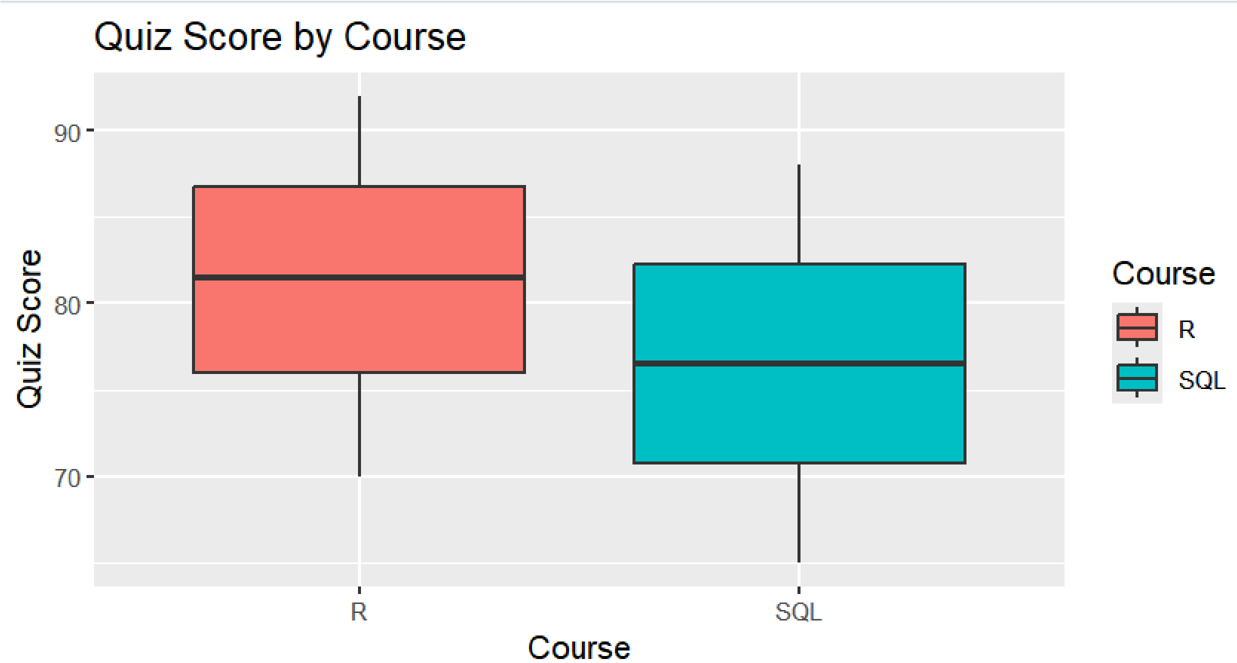


SET 9 – Online Learning Activity

Q1. Import CSV, Histogram & Boxplot

```
# Install packages (Run only once)

install.packages("ggplot2")

# Load package

library(ggplot2)

# Import CSV

data <- data.frame(

  Student_ID = c("L01","L02","L03","L04","L05","L06"),

  Gender = c("Male","Female","Male","Female","Male","Female"),

  Age = c(20,22,19,21,23,20),

  Course = c("R","R","SQL","R","R","SQL"),

  Study_Time = c(3.5,4.2,2.0,5.0,2.5,4.0),

  Videos_Watched = c(12,15,8,18,9,14),

  Quiz_Score = c(78,85,65,92,70,88),

  Login_Date = c("2025-01-05",
```

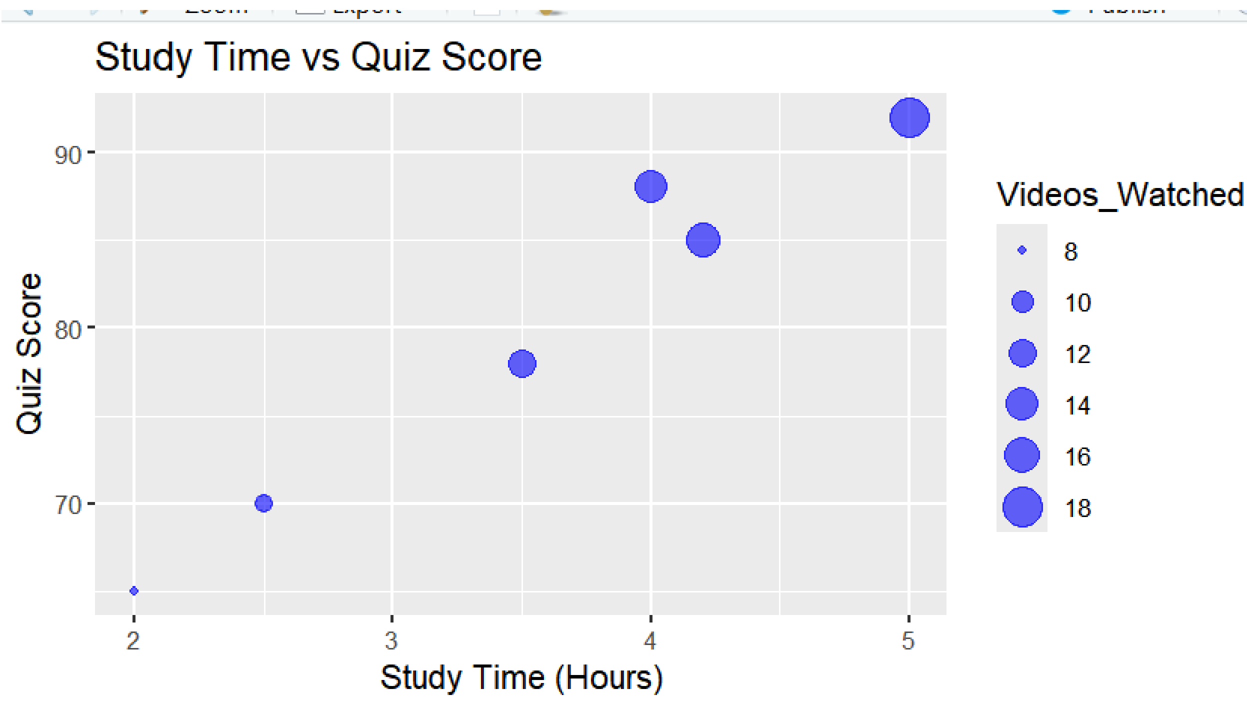
```r
        "2025-01-05",
        "2025-02-08",
        "2025-02-08",
        "2025-03-12",
        "2025-03-12")
)
# View data
print(data)

# Histogram
ggplot(data, aes(x = Quiz_Score)) +
  geom_histogram(binwidth = 5, fill = "skyblue", color = "black") +
  labs(title = "Histogram of Quiz Scores",
       x = "Quiz Score",
       y = "Frequency")

# Boxplot
ggplot(data, aes(x = Course, y = Quiz_Score, fill = Course)) +
  geom_boxplot() +
  labs(title = "Quiz Score by Course",
       x = "Course",
       y = "Quiz Score")
```



Q2. Scatter Plot (Bubble Chart)

```
library(ggplot2)

data <- data.frame(

  Student_ID = c("L01","L02","L03","L04","L05","L06"),

  Gender = c("Male","Female","Male","Female","Male","Female"),

  Age = c(20,22,19,21,23,20),

  Course = c("R","R","SQL","R","R","SQL"),

  Study_Time = c(3.5,4.2,2.0,5.0,2.5,4.0),

  Videos_Watched = c(12,15,8,18,9,14),

  Quiz_Score = c(78,85,65,92,70,88),

  Login_Date = c("2025-01-05",

               "2025-01-05",

               "2025-02-08",

               "2025-02-08",

               "2025-03-12",
```

```
          "2025-03-12")
)

ggplot(data,
    aes(x = Study_Time,
      y = Quiz_Score,
      size = Videos_Watched)) +
 geom_point(alpha = 0.6, color = "blue") +
 labs(title = "Study Time vs Quiz Score",
    x = "Study Time (Hours)",
    y = "Quiz Score")
```

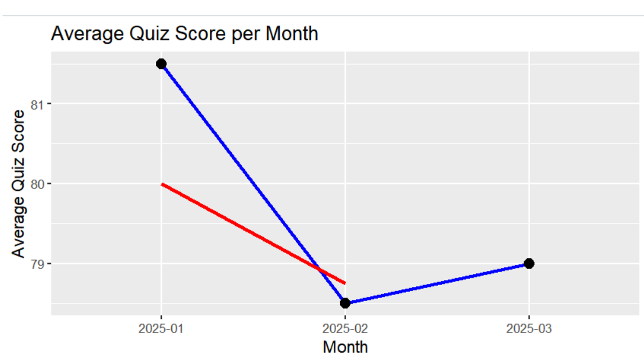


Q3. Date Conversion, Monthly Average & Line Chart

```
install.packages("dplyr")

install.packages("zoo")
```

```
library(dplyr)

library(ggplot2)

library(zoo)

data <- data.frame(

  Student_ID = c("L01","L02","L03","L04","L05","L06"),

  Gender = c("Male","Female","Male","Female","Male","Female"),

  Age = c(20,22,19,21,23,20),

  Course = c("R","R","SQL","R","R","SQL"),

  Study_Time = c(3.5,4.2,2.0,5.0,2.5,4.0),

  Videos_Watched = c(12,15,8,18,9,14),

  Quiz_Score = c(78,85,65,92,70,88),

  Login_Date = c("2025-01-05",

                 "2025-01-05",

                 "2025-02-08",

                 "2025-02-08",

                 "2025-03-12",

                 "2025-03-12")

)

# Convert to Date

data$Login_Date <- as.Date(data$Login_Date)

# Extract Month

data$Month <- format(data$Login_Date,"%Y-%m")

# Average Quiz Score

monthly <- data %>%

  group_by(Month) %>%
```

```
  summarise(Average_Score = mean(Quiz_Score))
print(monthly)
# Moving Average
monthly$Moving_Average <- rollmean(monthly$Average_Score,
                                   k = 2,
                                   fill = NA)
# Line Chart
ggplot(monthly,
       aes(x = Month,
           y = Average_Score,
           group = 1)) +
  geom_line(color = "blue", linewidth = 1) +
  geom_point(size = 3) +
  geom_line(aes(y = Moving_Average),
            color = "red",
            linewidth = 1) +
  labs(title = "Average Quiz Score per Month",
       x = "Month",
       y = "Average Quiz Score")
```



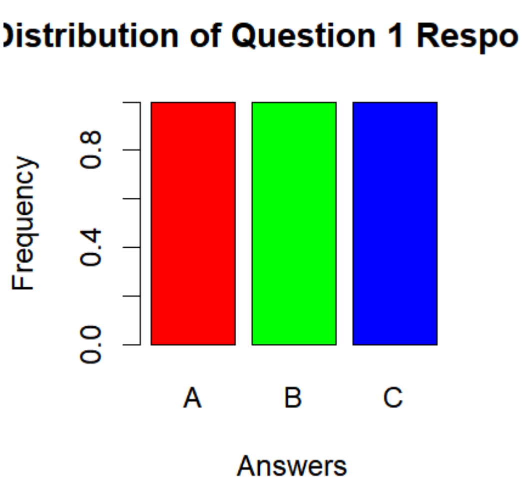
Q4. Tableau Dashboard

This question is performed in **Tableau**, not R.

SET 10 – Survey Responses Analysis

Q1. Grouped Bar Chart

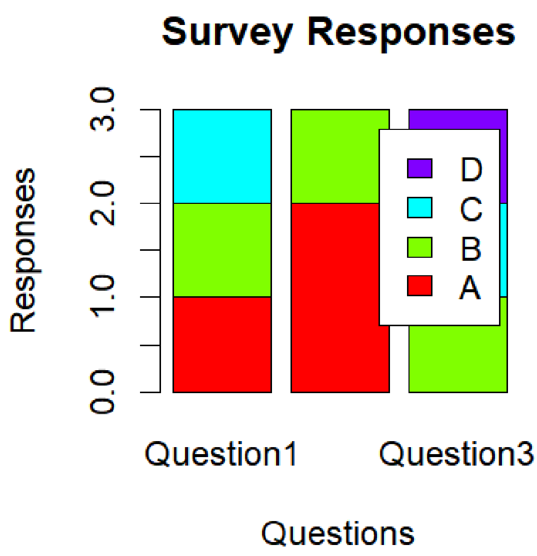
```
Survey_ID <- c(1,2,3)

Question1 <- c("A","B","C")


barplot(table(Question1),

     beside = TRUE,

     col = c("red","green","blue"),

     xlab = "Answers",

     ylab = "Frequency",

     main = "Distribution of Question 1 Responses")
```



Q2. Stacked Bar Chart

```
Question1 <- c("A","B","C")

Question2 <- c("B","A","A")
```

```
Question3 <- c("C","D","B")

responses <- rbind(
  table(factor(Question1, levels=c("A","B","C","D"))),
  table(factor(Question2, levels=c("A","B","C","D"))),
  table(factor(Question3, levels=c("A","B","C","D")))
)

rownames(responses) <- c("Question1","Question2","Question3")

barplot(t(responses),
        beside = FALSE,
        col = rainbow(4),
        legend.text = c("A","B","C","D"),
        xlab = "Questions",
        ylab = "Responses",
        main = "Survey Responses")
```



Q3. Table

```
Survey_ID <- c(1,2,3)


Question1 <- c("A","B","C")

Question2 <- c("B","A","A")

Question3 <- c("C","D","B")


survey_table <- data.frame(
  Survey_ID,
  Question1,
  Question2,
  Question3
)


print(survey_table)
```

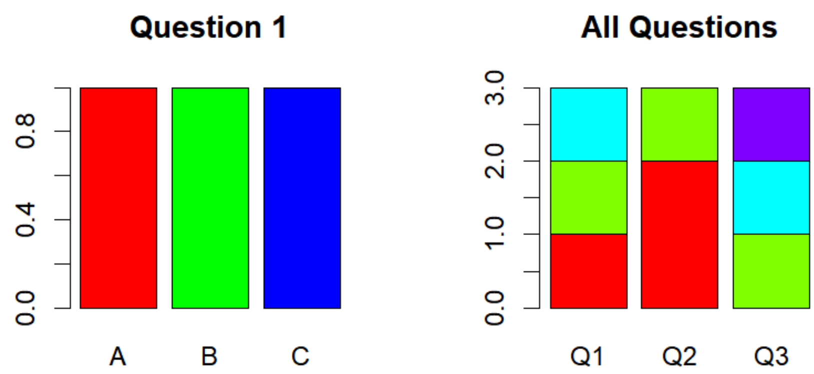
```
  Survey_ID Question1 Question2 Question3
1         1         A         B         C
2         2         B         A         D
3         3         C         A         B
>
```

Q4. Dashboard (R Equivalent)

```
par(mfrow = c(1,2))


Question1 <- c("A","B","C")


barplot(table(Question1),
        col = c("red","green","blue"),
        main = "Question 1")


Question2 <- c("B","A","A")
Question3 <- c("C","D","B")


responses <- rbind(
  table(factor(Question1, levels=c("A","B","C","D"))),
  table(factor(Question2, levels=c("A","B","C","D"))),
  table(factor(Question3, levels=c("A","B","C","D")))
)


rownames(responses) <- c("Q1","Q2","Q3")


barplot(t(responses),
        beside = FALSE,
```

col = rainbow(4),

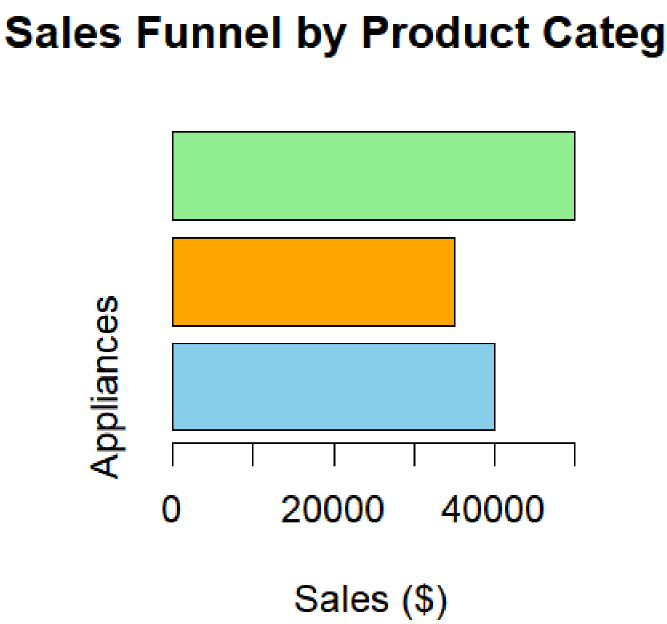
main = "All Questions")



SET 11 – Product Category Analysis

Q1. Funnel Chart (R Equivalent)

```
# Product Category Data

Category <- c("Electronics","Clothing","Appliances")

Sales <- c(50000,35000,40000)


# Funnel Chart (Horizontal Bar Chart)

barplot(rev(Sales),

    names.arg=rev(Category),

    horiz=TRUE,

    col=c("skyblue","orange","lightgreen"),

    xlab="Sales ($)",

    main="Sales Funnel by Product Category")
```



Q2. Table

```
Category <- c("Electronics","Clothing","Appliances")

Sales <- c(50000,35000,40000)


product_table <- data.frame(

  Category,

  Sales

)


print(product_table)
```

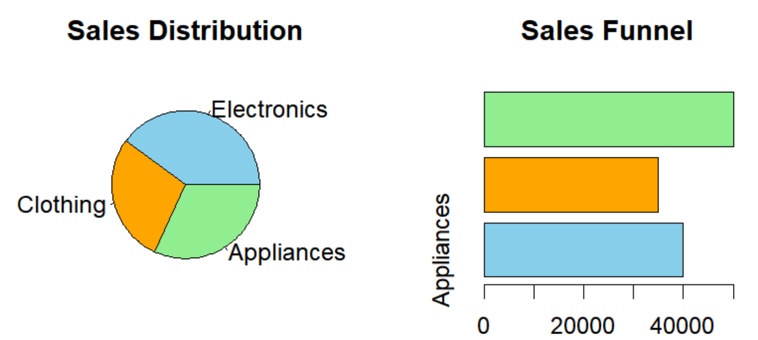
```
     Category Sales
1 Electronics 50000
2    Clothing 35000
3  Appliances 40000
>
```

Q3. Dashboard (R Equivalent)

par(mfrow=c(1,2))

Category <- c("Electronics","Clothing","Appliances")

Sales <- c(50000,35000,40000)



Pie Chart

pie(Sales,

labels=Category,

col=c("skyblue","orange","lightgreen"),

main="Sales Distribution")

Funnel (Horizontal Bar)

barplot(rev(Sales),

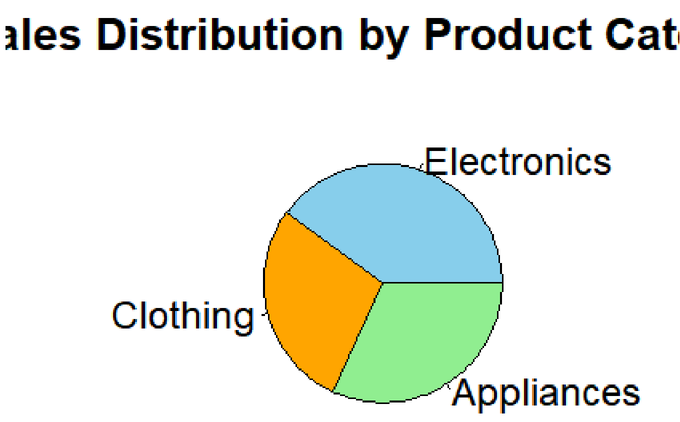
names.arg=rev(Category),

horiz=TRUE,

col=c("skyblue","orange","lightgreen"),

main="Sales Funnel")

Q4. Pie Chart



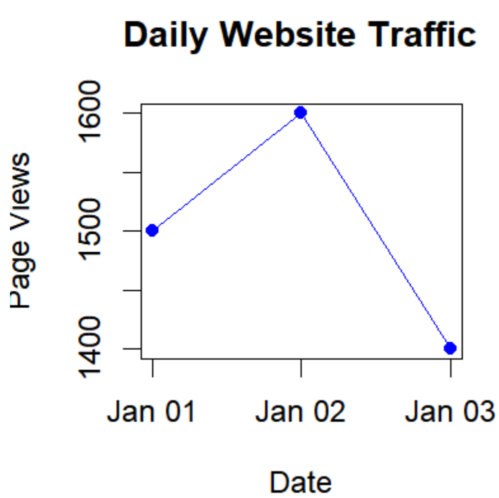

```
Category <- c("Electronics","Clothing","Appliances")

Sales <- c(50000,35000,40000)


pie(Sales,
   labels=Category,
   col=c("skyblue","orange","lightgreen"),
   main="Sales Distribution by Product Category")
```

SET 12 – Website Traffic

Q1. Line Chart

```
Date <- as.Date(c("2023-01-01",
             "2023-01-02",
             "2023-01-03"))


PageViews <- c(1500,1600,1400)
```

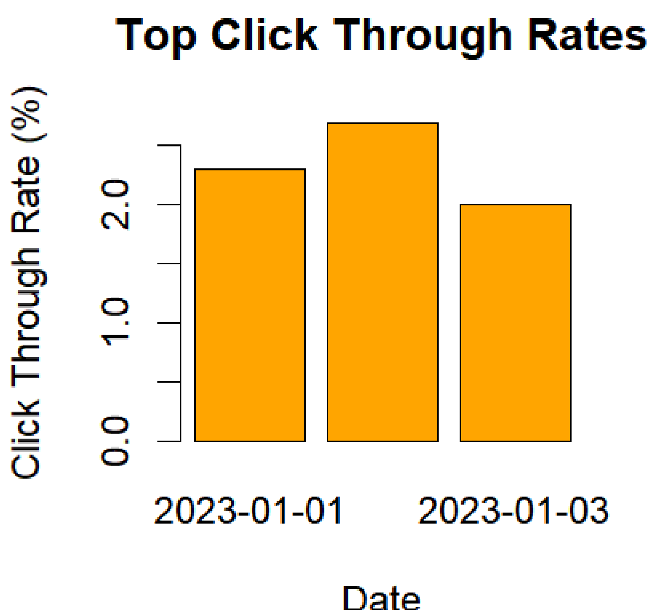
```
plot(Date,
    PageViews,
    type="o",
    pch=19,
    col="blue",
    xlab="Date",
    ylab="Page Views",
    main="Daily Website Traffic")
```



Q2. Bar Chart

```
Date <- c("2023-01-01",
        "2023-01-02",
        "2023-01-03")


CTR <- c(2.3,2.7,2.0)


barplot(CTR,
```

```
names.arg=Date,
col="orange",
xlab="Date",
ylab="Click Through Rate (%)",
main="Top Click Through Rates")
```



Q3. Table

```
Date <- c("2023-01-01",
     "2023-01-02",
     "2023-01-03")

PageViews <- c(1500,1600,1400)

CTR <- c(2.3,2.7,2.0)

traffic_table <- data.frame(
```

```
  Date,
  PageViews,
  CTR
)

print(traffic_table)
```

```
        Date PageViews CTR
1 2023-01-01      1500 2.3
2 2023-01-02      1600 2.7
3 2023-01-03      1400 2.0
>
```

Q4. Dashboard (R Equivalent)

```
par(mfrow=c(1,2))

Date <- as.Date(c("2023-01-01",
                  "2023-01-02",
                  "2023-01-03"))

PageViews <- c(1500,1600,1400)

plot(Date,
     PageViews,
     type="o",
     pch=19,
     col="blue",
     main="Page Views")
```

```
CTR <- c(2.3,2.7,2.0)

barplot(CTR,
    names.arg=c("Jan-01","Jan-02","Jan-03"),
    col="orange",
    main="Click Through Rate")
```